

To Train, or Not to Train: Exploring Foundation Models for TBAD Segmentation

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Abstract

Recent advancements in biomedical image analysis have been significantly influenced by vision foundation models (FMs) originally designed for general computer vision tasks. These models, such as the Segment Anything Model (SAM) have quickly transitioned into applications in medical image processing. This study evaluates three cutting-edge segmentation FMs using Type B Aortic Dissection (TBAD) CTA images. Our experiments critically assess their strengths and weaknesses compared to traditional methods, emphasizing their potential to enhance efficiency in medical image analysis. However, our findings also indicate that these FMs have not surpassed traditional segmentation models in terms of performance, suggesting limitations in their adaptability for medical image segmentation tasks straight out of the box.

Keywords: Medical Imaging, Foundation Model, Segmentation, Type B Aortic Dissection, CT Angiograms

1 Introduction

Medical image segmentation has made significant strides due to deep learning. However, developing and training segmentation models for new medical imaging (MI) data or tasks remains challenging due to the expensive and time-consuming nature of collecting and curating medical images, which often requires precise mask annotations from trained radiologists. These challenges could be alleviated with the advent of vision FMs. Recently introduced segmentation FMs, such as the Segment Anything Model (SAM) [Kirillov et al., 2023], have shown impressive performance across a range of segmentation tasks involving natural images.

In this study, we aim to empirically evaluate the performance of state-of-the-art (SOTA) segmentation FMs, specifically SAM, Segment Anything in Medical Images (MedSAM)[Ma et al., 2024] and Universal Medical Image Segmentation (UniverSeg)[Butoi et al., 2023]. By leveraging transfer learning from the pretraining phase, we assess the semantic segmentation predictions of these models on Computed Tomography Angiography (CTA) images. This approach aims to reduce the reliance on extensive annotations, streamlining the development and application of segmentation models in MI.

2 Related Works

2.1 Type B Aortic dissection (TBAD) segmentation

Type B aortic dissection (TBAD) is a rare but potentially fatal cardiovascular disorder characterized by a tear in the descending aorta. Precise and rapid segmentation of the true lumen (TL), false lumen (FL), and false lumen thrombosis (FLT) is essential for diagnosis and prognosis [Sailer et al., 2017].

2.2 Segmentation Foundation Models

SAM is a zero-shot segmentation foundation model designed to identify and segment specific objects within images using user-provided prompts. These prompts can include a single point, multiple points (including an entire mask), a bounding box (BB), or text. SAM has demonstrated exceptional performance across a wide range of natural image datasets. MedSAM is a fine-tuned version of SAM, trained on an extensive dataset

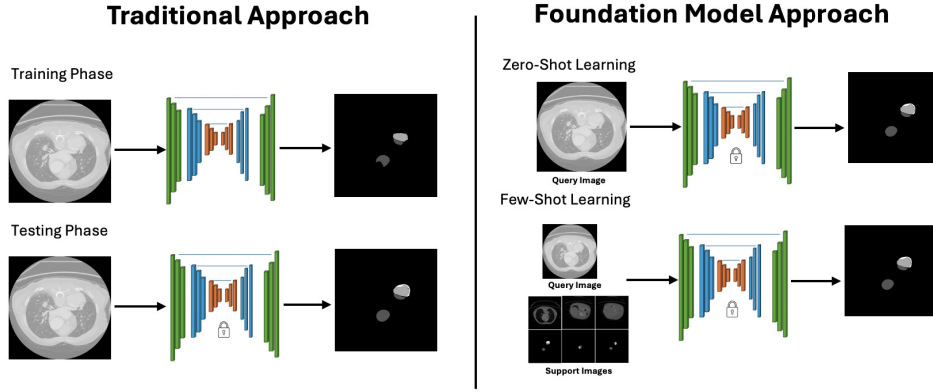


Figure 1: Traditional vs. FM Approach: In traditional segmentation, models like UNet are specifically trained to predict new images from scratch. Conversely, FMs leverage pre-trained models for inferring new tasks. Rather than retraining, these models use prompts like boundary boxes and support sets to make predictions

containing over one million pairs of medical images and corresponding masks. Empirical evidence consistently demonstrates that MedSAM outperforms existing state-of-the-art segmentation foundation models, frequently achieving performance levels comparable to or superior to specialized models in various studies.

UniverSeg is a few-shot FM designed for precise MI segmentation. It is trained on MegaMedical, a dataset comprising over 22,000 scans across various modalities. UniverSeg utilizes support sets of image-label pairs to effectively manage new tasks, allowing it to deliver robust performance on unfamiliar anatomies and tasks during evaluations.

3 Experiments

In this study, we utilized the ImageTBAD dataset [Yao et al., 2021], which includes 3D CTA images annotated with FL, TL and FLT. As part of our preprocessing steps, we transformed the 3D volumes into 2D images by considering each slice as a distinct image. Furthermore, for testing, we selected 628 images that contain all three lumens. The reason for this selection is that we wanted to evaluate the FMs capability to perform multi-class segmentation. By using only images that contain the region of interest, we ensure that the FMs provide meaningful, non-zero outputs for the specified queries and prompts.

Figure 2 showcases the predictions of all models. For MedSAM, the model is given an image and specific bounding box coordinates. In contrast, the UniverSeg model operates with a support set, using a prediction size of 64. During inference, it takes an image as input and generates a comprehensive multi-label mask, accurately identifying and labeling different segments within the image.

To empirically evaluate UniverSeg’s performance in testing CTA images, as described in the original paper, we experimented with different sizes for the support set "s", specifically 16, 32, and 64 2D image-label pairs. The support set was randomly selected from available images containing TL, FL, and FLT. During evaluation with MedSAM, each image from the testing set was processed individually by the model, resulting in separate masks generated for each BB. These masks for each lumen were then combined for evaluation metrics. Therefore, we focused solely on calculating the mean Intersection over Union (mIoU) score and mean Dice coefficient score (mDSC). As a benchmark, we used UNet model trained on nearly 23,000 images with various combinations of ROIs. The corresponding results are presented in table 1.

Table 1: Performance comparison of UniverSeg, MedSAM, and UNet on CTA image segmentation.

	MedSAM	UniverSeg			UNet
		s=16	s=32	s=64	
mIoU	0.070	0.180	0.219	0.188	0.632
mDSC	0.039	0.210	0.251	0.227	0.527

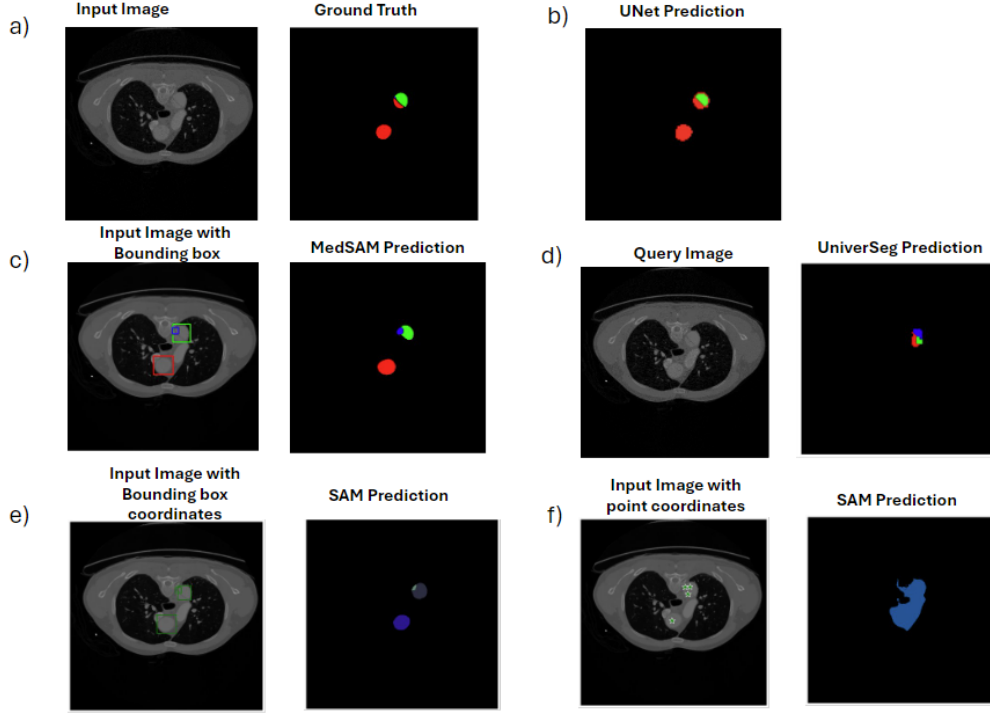


Figure 2: Comparison of input images, ground truth (GT), and predictions by UNet and three FMs. (a) Input image and GT. (b) Prediction by UNet. (c) Prediction by MedSAM using input images with bounding box (BB) coordinates. (d) Prediction by UniverSeg with a support set size of 64. (e) Prediction by SAM using BB coordinates. (f) Prediction by SAM using point coordinates. In the predicted masks and GT, blue regions represent FLT, red regions denote FL, and green regions correspond to TL.

4 Discussion

Our investigation into the performance of foundation models (FMs) trained on medical data and those trained on natural images but fine-tuned on medical data reveals that these models fail to outperform state-of-the-art (SOTA) segmentation models, such as UNet, on TBAD CTA images. It can be observed from figure 2 that the predicted mask from UNet is almost similar to the GT compared to other models. UniverSeg performs poorly on this particular image, which can be attributed to the method of selecting support size, which may not be optimal for all types of segmentation tasks. Conversely, MedSAM more accurately captures the shape of the lumen when provided with bounding box (BB) coordinates. SAM’s performance is comparable to MedSAM with BB coordinates but significantly declines when given point coordinates as input.

Several limitations of FM are highlighted in our study. SAM and MedSAM, for instance, are restricted to detecting only a single ROI per prompt, limiting their utility in complex MI scenarios. Furthermore, these models do not support multi-label segmentation, which is crucial for many MI tasks. They also necessitate prompts such as BB and point coordinates, requiring prior anatomical knowledge and professional expertise, making them less accessible for general use. Another significant limitation is that all evaluated FMs rely on 2D segmentation, despite the prevalence of 3D MI data in clinical practice, emphasizing the need for advanced models capable of handling 3D data.

Despite their limitations, models such as SAM and MedSAM, when fine-tuned for specific domains, show significant potential for improving clinical workflows through semi-automated annotation with user input. These models enable users to provide prompts to generate masks, which can be iteratively adjusted for accuracy.

Furthermore, generative models can be utilized to train these models on specific image modalities. To address the challenge of data scarcity, recent studies propose the generation and use of synthetic data [Liu et al., 2023, Abaid et al., 2024]. However, in some cases, 2D data alone is insufficient, particularly for video data or 3D volumetric data. As a future direction, we plan to fine-tune FMs using synthetic TBAD data and evaluate their performance in this context.

Moreover, recent research [Hooper et al., 2024] suggests that incorporating segmentation into classification tasks can enhance medical image classification, particularly with small datasets. This indicates a promising pathway for utilizing FMs to improve overall performance in medical imaging tasks. Future research should focus on integrating these models into classification workflows to leverage their capabilities in enhancing diagnostic accuracy and efficiency.

5 Conclusion

In conclusion, this paper explored the efficacy of SOTA segmentation FMs in the context of TBAD segmentation using CTA images. The study focused on evaluating models like SAM, MedSAM, and UniverSeg, leveraging their capabilities in MI segmentation through transfer learning. The results underscored the potential of these FMs to streamline the segmentation process by reducing the dependency on extensive manual annotations. Despite notable advantages in performance and adaptability shown by MedSAM and UniverSeg, for some MI modalities, limitations were identified. In essence, while there are challenges to overcome, the adoption of segmentation foundation models represents a significant step towards more efficient and accurate MI analysis, paving the way for enhanced healthcare delivery through advanced AI-driven technologies.

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